What is this thing, *determinism*?

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1 Introduction

- A word of thanks to Klaas Landsman

2 Introduction

- Brief overview of deterministic theories.
- Key questions to explore:
  - What are the defining features of a deterministic theory?
  - How does determinism relate to physical laws and initial conditions?
  - Are there examples of controversial or borderline deterministic theories?

3 Determinism: attractive and repulsive

- Philosophers have been repulsed by determinism because of its consequences for human action and moral responsibility
- Physicists have been attracted by determinism because of ...

4 Philosophy against natural philosophy

- Compatibilism might actually be true. But I suspect that it is also motivated by anti-empiricism.
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5 Determinism: from world to theory, and back again

- Carnap
- Lewis

6 Possible worlds analyses

- Earman, Primer of Determinism

7 Melia's distinction

8 Definition

Definition: We say that a theory T is **Landsman-deterministic** just in case for any two models W and W' of T, and initial segment $U \subseteq W$, if $f : W \to W'$ and $g : W \to W'$ are isomorphisms such that $f|_U = g|_U$, then $f = g$.

9 What is a theory?

10 Ideas and Concepts

- Theoretical Definitions
  - Notes on definitions of determinism in literature.
- Connections to Physical Theories
  - Examples: Newtonian mechanics, quantum mechanics, general relativity.

11 PDF Annotations

- Below are links to PDFs for reference. Use org-noter to annotate them.

** Key References

- nil
- nil
- nil
- nil
- Halvorson, Closing the hole argument
- Landsman, Re-opening the hole argument
- nil
- Recent stuff by Pooley, Read, etc.?

- Menon and Read, Some remarks on recent formalist responses to the hole argument nil
- nil
- `https://philpapers.org/browse/determinism`
- nil
- nil
- nil

** Additional Resources

- Additional Reading 1
  - Notes:
- Additional Reading 2
  - Notes:

12 Questions for Discussion

- What is the role of initial conditions in deterministic theories?
- Can a theory be deterministic at one level and indeterministic at another?
- How does determinism connect with predictability?

13 To-Do

☐ Read and annotate PDFs using org-noter.

☐ Summarize main points from annotations.

☐ Draft a section on examples of deterministic theories.

14 Notes

- Add any additional ideas, thoughts, or observations here.